

09/03/26

1 slip → resources → answers
plots, gradient descent
plotting

2016

Sig KDD

* XG Boost :

- Extreme Gradient Boosting
[XG]

ensemble algorithm

GBM → Gradient Boosting Machines

for large datasets

- large datasets
- tabular data

Example

ABDM

↳ Ayush Bharat Digital Mission

↓
optimization

Algorithm
level
optimization

System level
optimization
- parallel running

- Regularization added to a decision tree

- column subsampling and shrinkage ⇒ feature random subset in random forest

↳ the weight parameter in bagging

- split finding algorithms

exponential
computationally
by experience

- Basic exact greedy algorithm

- approx. algorithm using weighted quantile sketch

- histogram based method

from quantile
pick up the split
points

- sparsity aware split finding
(handling missing values)

binning

• splitting - aware split linking

	Dosage	Log effectiveness	Residuals
S ₁	10	-7	-7.5
S ₂	999	-3	-3.5
S ₃	21	7	6.5
S ₄	25	8	7.5
S ₅	5	-5	-5.5
S ₆	792	-2	-2.5

split the dataset by samples which have non-numerical values 999

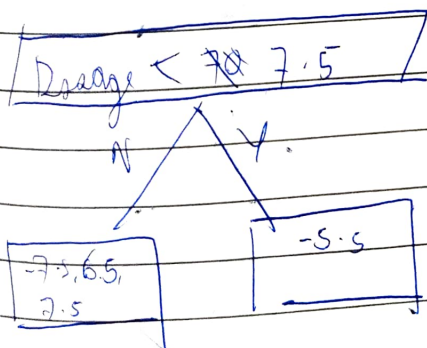
Dosage	Residuals
S ₁	-5.5
10	-7.5
21	6.5
25	7.5

Dosage	EFF	Residual
999	-3	-3.5
990	-2	-2.5

$$IG_{right} = \{-5.5, -3.5, -2.5\}$$

① log odds ② Probabilities

η_1	η_2	η_3	y	
-	-	-	Yes	=
-	-	-	Yes	
-	-	-	No	
-	-	-	Yes	



$$0.3 \rightarrow 0.7$$

or

$$1 - 0.7 = 0.3$$

$$0 - 0.4 = -0.4$$

the values can be calculated

$$IG_{left} = \{-7.5, 6.5, 7.5, -3.5, -2.5, -5.5\}$$

higher IG that optimum split path

residual

Note as is limit
If more samples are there ~~xxxx~~

why default path

$$IG_{left} > IG_{right}$$

Fitting to the data

Move the missing values to both the end and then whichever gives higher information gain we choose that.

if I don't have values only

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* System Design Improvements :-

* Parallel Learning :-

sample sizes

eg :-

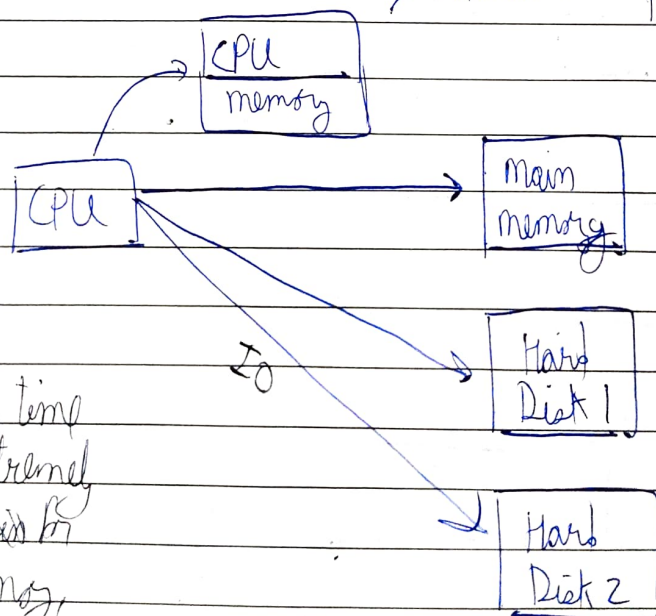
1	2
3	4

x_1
5
28
6
54
...
million



For sorting this big list, we distribute the values to different computers and at last the aggregated sorted values are returned.

* Cache - Aware Access :-



IO access time is extremely high in Main memory, hard disk.

IO access time
CPU memory < Main memory < Hard Disk

~~giga~~ XGBoss will store more frequently required data as cache for easy access

each CPU
targets a distributed
set of data

Then decompressed multiple CPUs
multiple cores
Then aggregated

classmate
Date _____
Page _____

Blocks for out of core computation:

Sharding:

Distribute data into clusters of databases

* ML -
Storing data in compressed form amongst different CPUs.

* LightGBM - performs better than XGBoost
↓ ↓
2017 2016

XGBoost - hist → library
in

* Gradient Boosting for Classification:

skit-learn

	Chest Pain	Good blood circulation	Blocked Arteries	Heart Disease
S1	No	No	No	No 0
S2	Yes	Yes	Yes	Yes 1
S3	Yes	Yes	No	Yes 1
S4	Yes	No	No	Yes 1
S5	Yes	No	Yes	Yes 1
S6	No	Yes	No	No 0

① log- odds

② convert to prob

$f_0 \neq 0$ ~~was~~ but here

$y_{obs} = 0$ & can't be

even $0 = 0$ or $1 \neq 0$ can't
be because it is encoding

* Step 1: Initial Prediction

$$\log(\text{odds}) = \log\left(\frac{1}{2}\right) \approx -0.693 \approx \boxed{0.7}$$

↓
No Yes

odds: $\frac{\text{no. of success}}{\text{no. of failures}}$

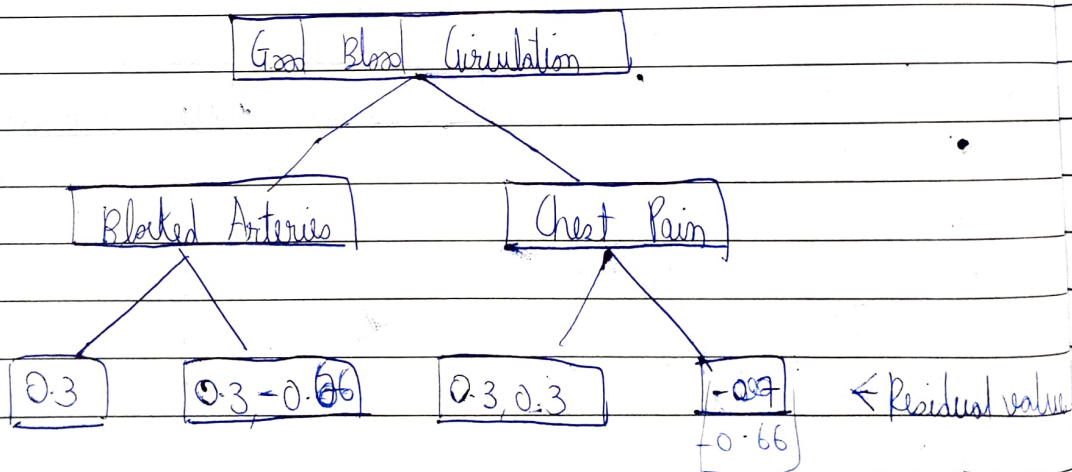
2) Convert this: to probability

$$p = \frac{e^{\log(\text{odds})}}{1 + e^{\log(\text{odds})}}$$

$$\Rightarrow \frac{e^{\log(\frac{1}{2})}}{1 + e^{\log(\frac{1}{2})}}$$

$$p \Rightarrow \underline{\underline{0.66}}$$

Heart Disease		Residual	
No	0	0-0.66	-0.66
Yes	1	1-0.66	0.33
Yes	1	1-0.66	0.33
Yes	1	1-0.66	0.33
Yes	1	1-0.66	0.33
No	0	0-0.66	-0.66



Σ Residuals

probabilities

Step 3:

natural

Predict the next residuals & convert the leaf values to log(odds)

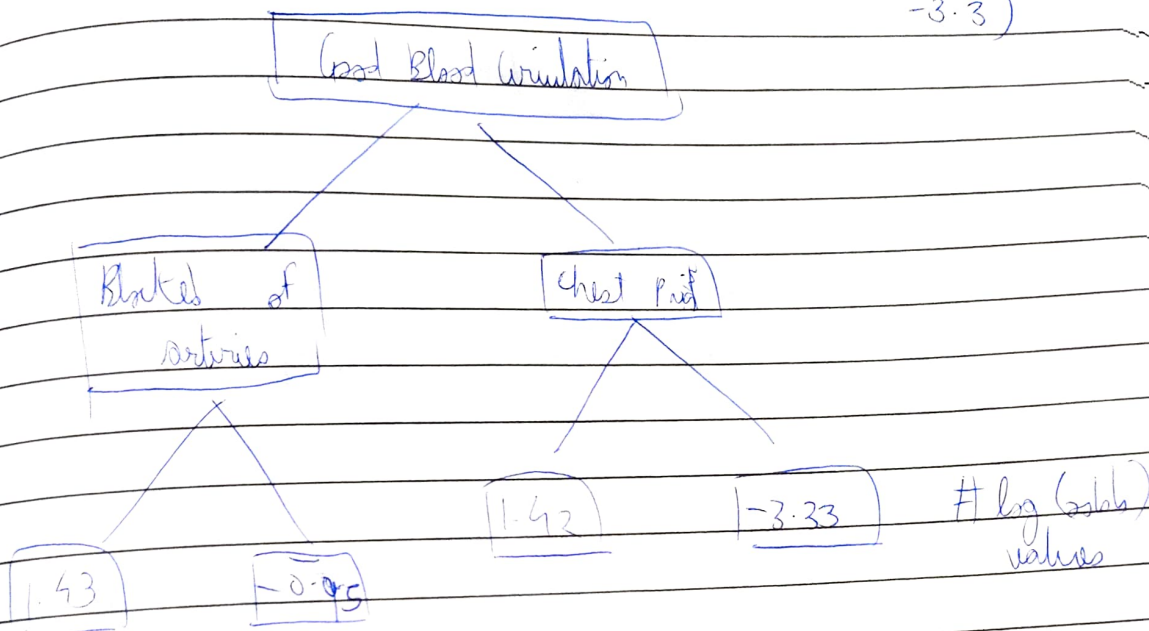
log odds

Σ Residuals

sample/value is each

for 1st leaf $\Rightarrow \frac{0.3}{0.66 \times (1 - 0.66)} \approx \underline{\underline{1.53}}$

for 2nd leaf $\rightarrow 0.95 \rightarrow \text{answer}$, 3rd leaf, value -1.42 , 4th leaf -3.33



- Then now calculate a probability value by converting log (odds) to probability increase
- Take only created 1 model ~~decision tree~~ for case, we will create many other model.
- Then we will see if it is ^{lower} than 0.5 $\leftarrow$ if lower, the model is ~~select~~ ^{prob. (weight failure)} and above means ~~success~~